A DEPTH LOWER BOUND OVER A COHEN-MACAULAY CLOSED FIBRE OF A FLAT LOCAL HOMOMORPHISM

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- The lifting of regular sequences along a flat local homomorphism from the closed fibre. Invoked at the start of the counted proof (source lines 2105-2107). Lets a regular sequence detected on the closed fibre be used in the ring itself, which reduces the statement to the case of a field.
- The depth and dimension inequality for ideals in a Cohen-Macaulay ring. Invoked at the end of the counted proof (source line 2108). Supplies the inequality in the field case to which the proof reduces.

Source reference. Original reference algebra-lemma-grothendieck-regular-sequence: sources/stacks-dualizing/pinned-original-source.tex, source lines 2105–2105. Navigation pointer only: the counted text refers to the Stacks item algebra-lemma-grothendieck-regular-sequence, which belongs to another chapter of the project and is not part of this card. The pointer carries the target's own label as its number because this chapter's staged snapshot does not contain the other chapter's tag table; its content is not reproduced here.

Source reference. Original lemma lemma-depth-CM: sources/stacks-dualizing/pinned-original-source.tex, source lines 2061–2061. Navigation pointer only: the counted text refers to the Stacks item lemma-depth-CM, which is an existing labeled result of this same chapter and is not part of this card. The pointer carries the target's own label as its number and its definition line as its span; its content is not reproduced here.

Lemma 0.1. *Let $R \to S$ be a flat local ring homomorphism of Noetherian local rings. Denote $\mathfrak{m} \subset R$ the maximal ideal. Let $I \subset S$ be an ideal. If $S/\mathfrak{m}S$ is*

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Cohen-Macaulay, then

$$depth_I(S) \geq \dim(S/\mathfrak{m}S) - \dim(S/\mathfrak{m}S + I)$$

Proof. By Algebra, Lemma algebra-lemma-grothendieck-regular-sequence any sequence in S which maps to a regular sequence in $S/\mathfrak{m}S$ is a regular sequence in S. Thus it suffices to prove the lemma in case R is a field. This is a special case of Lemma lemma-depth-CM. □